



Metaplastic breast carcinoma: more than a special type

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Abstract | Metaplastic breast carcinoma (MBC) is a special histological type of invasive breast cancer. MBC is a descriptive and operational term for a heterogeneous collection of tumours with distinct histologies, clinical behaviours and potentially responses to therapy.

Metaplastic breast carcinoma (MBC), which accounts for 0.2–5% of all invasive breast cancers, is an umbrella term for a heterogeneous collection of cancers that are characterized by the presence of neoplastic cells displaying differentiation towards squamous epithelium or mesenchymal components, such as spindle, chondroid, osseous or rhabdoid cells¹ (FIG. 1). Given the panoply of histological appearances, several synonyms for MBCs have been used in the literature, including carcinosarcoma, sarcomatoid carcinoma, carcinoma with sarcomatous stroma, matrix-producing carcinoma, adenosquamous carcinoma and fibromatosis-like metaplastic tumour, to name but a few¹. Although it is currently unknown whether these heterogeneous lesions have similar biological characteristics and clinical behaviours, they are perceived clinically as a single entity¹. As a group, MBCs (>90%) display a triple-negative phenotype (that is, they lack expression of oestrogen receptor (ER), progesterone receptor (PR) and ERBB2 (also known as HER2)), preferentially have a basal-like or claudin-low molecular subtype and frequently harbour mutations in the *TP53* gene (which encodes p53)^{2–4}. Although MBCs display many similarities with common-type triple-negative breast cancers (TNBCs), several questions remain germane to our understanding of MBCs. Why do they display divergent differentiation? What is the cell of origin of these tumours? And why do patients with MBC seem to less frequently respond to chemotherapy than those with other types of TNBCs?

Several studies have sought to define whether breast cancers of different subtypes stem from the transformation of different cell lineages in the mammary gland or cells at different differentiation stages⁵. In the case of MBCs, it was initially thought that these cancers might originate from stem cells or myoepithelial cells, given that these cells would be able to display the differentiation patterns observed in MBCs. Conditional mouse model studies, however, have demonstrated that inactivation of *Trp53* and *Brca1* in the luminal progenitor cells and basal cells

of the mouse mammary gland often, but not exclusively, result in the development of mammary tumours that display the cardinal histological characteristics of MBCs^{6,7}. These observations imply that although the phenotype of a tumour can be strongly influenced by the phenotype of its cell of origin⁵, depending on the founder genetic aberrations, the transformation of different cell lineages might result in the development of MBCs, an observation suggestive of a convergent phenotype.

Another potential mechanism for the existence of multiple phenotypic patterns in MBCs is epithelial-to-mesenchymal transition (EMT)^{2,4}. It is unclear whether the EMT observed in MBCs arises as a result of the phenotypic plasticity of its cell of origin or as a result of epigenetic or somatic genetic aberrations that affect genes that directly or indirectly control EMT. Although mutations that affect components of the WNT pathway, which has been shown to have a role in EMT in cancers, were initially reported in MBCs⁸, subsequent studies^{2,9} failed to identify mutations in these genes.

MBCs are generally characterized by high levels of genetic instability and harbour gene copy number alterations that are similar to those found in common-type TNBCs². Currently, information on the mutational landscape of MBCs is scarce: no pathognomonic mutation or recurrent fusion gene that accounts for the histological characteristics of these cancers has been identified, and it is unknown whether different repertoires of somatic mutations drive the distinct subtypes of MBCs. The morphologically and phenotypically distinct components of individual MBCs have been shown to be clonal on the basis of the presence of identical *TP53* mutations¹⁰. However, intra-tumour genetic heterogeneity, as defined by differences in the somatic genetic alterations between phenotypically distinct components of individual cases, is also evident¹⁰. Mutations in *TP53* and phosphatidylinositol-4,5-bisphosphate 3-kinase, catalytic subunit alpha (*PIK3CA*), and losses of PTEN and cyclin-dependent kinase inhibitor 2A (*CDKN2A*) have

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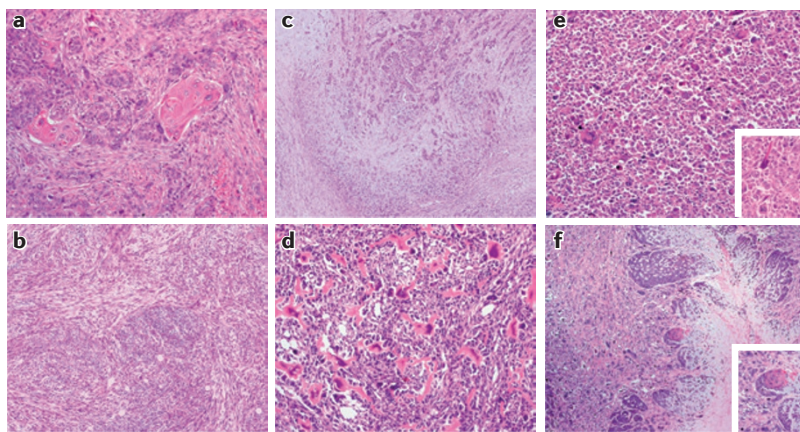


Figure 1 | The phenotypic diversity of metaplastic breast cancers. Metaplastic breast cancers can display differentiation towards squamous epithelium (part **a**), spindle (part **b**), chondroid matrix (part **c**), osseous (part **d**) and rhabdoid (part **e** and inset) cells. Often, however, multiple lineages of differentiation can be found in the same tumour, including squamous, spindle, chondroid and ductal (part **f** and inset) cells.

been recurrently found in MBCs; however, they do not seem to be restricted to tumours of specific phenotypes^{2,10} and are not unique to MBCs. Overexpression and amplification of the epidermal growth factor receptor (*EGFR*) gene have been reported in a subset of MBCs and seem to be more prevalent in tumours with squamous and/or spindle cell metaplasia¹¹.

Based on the phenotypic and genetic similarities between sporadic TNBCs and breast cancers arising in *BRCA1* germline mutation carriers, it has been hypothesized that *BRCA1* pathway dysfunction might also have a role in sporadic TNBCs¹². Although it is only present in a minority of sporadic triple-negative invasive ductal carcinomas of no special type (IDC-NSTs), methylation of the *BRCA1* gene promoter has been reported in approximately two-thirds of MBCs¹². The biological impact of *BRCA1* promoter methylation, however, remains to be fully determined. Although some argue that this epigenetic phenomenon might result in impaired homologous recombination-based repair of DNA double-strand breaks and sensitivity to platinum salts and poly(ADP-ribose) polymerase (PARP) inhibitors, others have argued that in MBCs, *BRCA1* promoter methylation could be merely reflective of its cell of origin or differentiation status¹³. Consistent with this idea, MBCs have been shown to be relatively resistant to DNA-damaging chemotherapy agents compared to TNBCs.

A better understanding of the molecular features of MBC, in particular the constellation of somatic mutations that underpin the different phenotypes, is germane to identifying novel therapeutic strategies for patients with MBC. In such studies, the classification of MBCs into subtypes will be essential, as will be the separation of their distinct morphological components prior to genetic analyses. *PIK3CA* mutations seem to be more common in MBCs than in other TNBCs²; however, further investigation is required to demonstrate whether these genetic aberrations constitute predictors of sensitivity to PI3K-mTOR pathway inhibitors. Likewise, *EGFR* gene amplification reported in a subset of MBCs has yet to be formally

tested as a predictor of response to anti-*EGFR* therapies in these tumours. It should be noted, however, that given the frequent genetic and epigenetic aberrations that affect *PTEN* in MBCs, which might result in resistance to inhibition of upstream tyrosine kinase receptors, it is unlikely that these cancers will display an overwhelming response to these agents.

Another fascinating yet relatively unexplored issue is the fact that MBCs might constitute a convergent phenotype (the end product of distinct combinations of genetic and/or epigenetic aberrations). Consistent with this concept is the fact that both high-grade squamous cell carcinomas and myoepithelial carcinomas can have areas of overt spindle cell morphology and, when predominantly composed of spindle cells, they are currently classified as MBCs with spindle cell metaplasia¹. Ascertaining whether these distinct types of spindle cell carcinomas are underpinned by similar or different genetic and epigenetic aberrations will be crucial for our understanding of the pathogenesis of these lesions and their classification.

Future studies investigating whether distinct subtypes of MBC or even distinct phenotypic components within a given MBC are underpinned by different repertoires of driver genetic and/or epigenetic aberrations might not only provide insights into the biology of these neoplasms but also reveal the basis of their heterogeneity. Such endeavours will be instrumental for the identification of novel therapeutic targets and for the development of rational combinations of targeted therapies to realize the potential of precision medicine for patients with MBC.

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Competing interests statement

The authors declare no competing interests.